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Convex Optimization for Bundle Size Pricing Problem

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Abstract. We study the bundle size pricing (BSP) problem in which a monopolist sells bundles of products to customers and the price of each bundle depends only on the size (number of items) of the bundle. Although this pricing mechanism is attractive in practice, finding optimal bundle prices is difficult because it involves characterizing distributions of the maximum partial sums of order statistics. In this paper, we propose to solve the BSP problem under a discrete choice model using only the first and second moments of customer valuations. Correlations between valuations of bundles are captured by the covariance matrix. We show that the BSP problem under this model is convex and can be efficiently solved using off-the-shelf solvers. Our approach is flexible in optimizing prices for any given bundle size. Numerical results show that it performs very well compared with state-of-the-art heuristics. This provides a unified and efficient approach to solve the BSP problem under various distributions and dimensions.

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1. Introduction

We study the bundle pricing problem of a risk-neutral monopolist selling a set of heterogeneous products. The pricing mechanism we consider is bundle-size pricing (BSP) by which the firm offers a menu of bundles with different sizes and prices. The price of each bundle depends only on the size, that is, the number of items within the bundle. Customers have random valuations on the products and purchase bundles to maximize their surplus. The objective of the firm is setting bundle prices to maximize the expected profit.

Among several alternative bundling mechanisms, BSP seeks a balance between two extreme mechanisms: one is *mixed bundling*, in which firms bundle every possible combination of products with a price; the other is *pure bundling*, in which only a grand bundle is offered to customers. Clearly, mixed bundling enjoys the highest flexibility but is seldom adopted in practice because of the complexity in managing exponentially many different bundles. Pure bundling is simple enough but does not allow customers to choose their preferred products accordingly. In BSP, firms only need to set a few prices for bundles, and customers can flexibly customize their own choices. It is also shown that BSP can approximate mixed bundling very well in yielding profits in many cases (Chu et al. 2011).

In practice, BSP is commonly adopted in several industries. For example, cable TV companies allow customers to choose combinations of channels within the prescribed cardinalities of bundles (Wu et al. 2019). Theater companies and sports teams sell bundles of tickets of different games and plays to customers in a season, and the price of each bundle typically depends only on the size of the bundle (Chu et al. 2011). Streaming service provider Netflix offers three plans based on the number of devices on which one can stream Netflix at the same time (Netflix 2019); basic, standard, and premium plans allow one, two, and four devices, respectively.

Despite the simplicity and profitability of BSP in practice, efficiently finding optimal prices for the BSP problem is extremely hard because it involves optimizing over order statistics (Wu et al. 2008, Abdallah et al. 2021). Specifically, for given sizes of bundles, customers first choose the surplus-maximizing products for each size of bundle and then select the surplus-maximizing bundle over all sizes. For given prices, identifying the distribution of customer valuations over bundles is equivalent to characterizing the distribution of the maximum sums over order statistics, which is a complicated task. Even if we can characterize the distribution, it is still unknown whether the resulting pricing problem is tractable. Thus, an efficient approximation scheme with good performance is needed.

In BSP, customers' surplus maximization is essentially a random choice problem, and thus, it is natural to construct approximations using discrete choice modeling. Along this way, it is important to note that the valuations of different sizes of bundles are correlated and capturing such correlation structure is crucial for constructing good approximations (Chu et al. 2011). However, commonly used parametric choice models in the literature are either unable to capture the interdependency adequately or lead to an intractable pricing problem, for example, the *mixed logit model* (Li et al. 2019). Although some other choice models, such as the *nested logit* or *multinomial probit* (MNP), can capture correlations, it is unknown how to specify the nests for the former model and whether the resulting pricing problem is tractable under the latter model.

In this paper, we study the BSP problem under a discrete choice model called the "cross-moment model" (CMM). This model seeks an extremal distribution from a set of distributions with a fixed mean vector and covariance matrix to maximize customers' utility surplus. Correlations of valuations over bundles are captured by the given cross moments, which can be easily simulated given the distributions of valuations over products. Our main contributions are listed as follows:

- We show that the BSP problem under CMM is convex and efficiently solvable using off-the-shelf solvers. To the best of our knowledge, we are the first to solve the BSP problem using convex optimization techniques. This tractability result combined with the distribution-free nature of CMM give rise to a unified and efficient approach for solving the BSP problem under various distributions and dimensions.
- We examine the performance of CMM via extensive simulation studies. Although it is an approximation of the original BSP problem, numerical results show that the approximation error of CMM is much smaller than the approach that fails to capture correlations. Moreover, we show that CMM is very competitive to the simulation optimization method, which is the state-of-the-art heuristic for small-scale BSP problems in the literature. We also demonstrate that CMM performs very well compared with other approximation schemes for both small- and large-scale problems.

The technique proposed in this paper can also be extended to other classes of pricing problems, such as mixed bundling, as long as they involve a discrete choice constraint (i.e., one out of n options) and the (random) utilities of the options are correlated and must be explicitly incorporated in the choice model. This should allow the technique to appeal to a wider community.

1.1. Organization

The paper is structured as follows: In Section 2, we review the related literature. In Section 3, we formally define the bundle size pricing problem. In Section 4, we first propose to solve the BSP problem under the cross-moment model; we then derive a reformulation with choice probabilities as decision variables and establish the convexity of the reformulation. The numerical experiment is presented in Section 5, and finally, this paper is concluded in Section 6.

1.2. Notations

Throughout the paper, scalars are denoted by standard letters, such as x; vectors are denoted by bold letters, such as $\boldsymbol{x}$; and matrices are denoted by bold capital letters, such as $\boldsymbol{X}$. We denote the transpose of a column vector $\boldsymbol{x}$ and a matrix $\boldsymbol{X}$ as $\boldsymbol{x}^T$ and $\boldsymbol{X}^T$, respectively. We also use $\boldsymbol{e}$ to denote the vector of all ones and denote $\boldsymbol{e}_i$ the vector with one in the ith entry and zeros elsewhere. Random variables are denoted by letters with tilde notation, such as $\tilde{u}$ and $\tilde{\boldsymbol{u}}$. We denote $\mathrm{Diag}(\boldsymbol{x})$ the diagonal matrix with the entries of $\boldsymbol{x}$ along the diagonal and denote $\mathrm{diag}(\boldsymbol{X})$ the vector formed by the diagonal entries of $\boldsymbol{X}$. Finally, $\mathrm{tr}(\boldsymbol{X})$ is the trace of $\boldsymbol{X}$.

2. Literature Review

Our work is mainly related to three streams of literature.

2.1. Bundle Size Pricing

Bundling mechanisms have been studied extensively for a long time. Earlier notable papers include Stigler (1963), Adams and Yellen (1976), and McAfee et al. (1989), which focus on the benefits and limitations of bundling but are restricted to the two-product case. Recent studies include Bakos and Brynjolfsson (1999), Chen et al. (2017), and Abdallah (2019), which focus on large-scale pure bundling. However, the literature on BSP is relatively sparse although it is an attractive alternative to other common mechanisms. The concept of BSP is first proposed by Spence (1980), who shows the efficacy of quantity-based pricing on extracting consumer surplus. Hitt and Chen (2005) investigate the conditions under which mixed bundling can be reduced to so-called "customized bundling," which is essentially the BSP problem.

A notable study of BSP is Chu et al. (2011), who show by extensive simulations that BSP tends to closely approximate the profits from mixed bundling. They solve the BSP problem using the simulation-optimization method. Recently, Abdallah et al. (2021) analyze the asymptotic properties of the large-scale BSP and provide guidance on how to construct closed-form pricing policies. Ma and Simchi-Levi (2021) propose a type of simple mechanism termed "pure bundling with disposal for cost" and prove performance guarantees for this mechanism under different settings.

Despite the rich literature on understanding different bundling mechanisms, little is known on how to efficiently compute optimal bundle prices of BSP. In addition to Chu et al. (2011) and Abdallah et al. (2021), who develop different approximation schemes, Hitt and Chen (2005) propose a combinatorial technique to solve the BSP problem under the assumption of the Spence–Mirrlees single-crossing property of customer valuations. Under the same assumption, Wu et al. (2019) extend Hitt and Chen (2005) to explore the structural properties of this problem. They reformulate the BSP as a linear program and give a combinatorial algorithm. Wu et al. (2008) develop a nonlinear mixed integer program for the BSP problem under deterministic valuations and solve the model using heuristics.

These computational approaches either assume deterministic customer valuations and build integer programs, which are difficult to solve, or impose special structures. In this paper, we contribute to this literature by relaxing all these assumptions and developing a convex optimization framework. Our approach allows stochastic customer valuations with arbitrary correlations. More importantly, our approach is flexible to optimize any given bundle sizes with efficient computation and good numerical performance.

2.2. Multiproduct Pricing Under Choice Models

Our work closely relates to the rich literature on the multiproduct pricing problem under discrete choice models. A key question to answer along this line of literature is whether the pricing problem is convex under a particular class of choice models. An early negative result is by Hanson and Martin (1996), who find that the profit function of the multiproduct pricing problem under the MNL model is nonconcave with respect to pricing decisions.

However, Xue and Song (2007) and Dong et al. (2009) show that, although the pricing problem is nonconcave in prices, it is indeed concave in choice probabilities (market shares) through a reformulation. This interesting finding is extended to other choice models such as nested logit (Li and Huh 2011), d-level nested logit (Huh and Li 2015, Li et al. 2015), paired combinatorial logit (Li and Webster 2017), and more recently to the generalized extreme value family (Zhang et al. 2018). In more general settings, several studies report negative results. For example, Gallego and Wang (2014) show that the convexity result fails to hold for the nested logit model under product-differentiated price sensitivities. Li et al. (2019) show that the pricing problem under the mixed logit model is neither concave nor quasi-concave.

Our work is based on the CMM proposed by Mishra et al. (2012). This model belongs to the semiparametric choice model (Natarajan et al. 2009, Feng et al. 2017), which assumes that the distribution of random utility is restricted in a prescribed set of distributions, and only partial information is revealed to the modeler. CMM assumes that the modeler has access to the cross-moment information. Choice probabilities of CMM are given by the optimal solution to a semidefinite program (SDP). Recently, Ahipaşaoğlu et al. (2015) apply CMM to traffic assignment problems. Qi et al. (2019) study product line optimization using a marginal moment model, which is another subclass of semiparametric choice model. We contribute to this literature by applying CMM to solve the BSP problem.

2.3. Pricing with Moments

Pricing with moment information is not new. There is a rich literature on robust pricing models (see, e.g., Bergemann and Schlag 2011, Carroll 2017, Chen et al. 2017). We refer the readers to Chen et al. (2017) and the references therein for more details on this literature. Similar to our model, this literature assumes that the true distribution of customer valuation (type) is unknown although partial information, such as support, moment, and marginal distribution, are assumed given. The firm then makes decisions to maximize the worst case (minimum) profit or minimize the worst case (maximum) regret over all feasible distributions.

Most models in this literature focus on distributions attaining the worst-case performance in terms of the seller's objective, while our model (CMM) focuses on a distribution that attains the best-case performance for the buyer and uses it as a reference to price the bundles. Extensive numerical results from Mishra et al. (2012), Ahipaşaoğlu et al. (2015), and Natarajan and Teo (2017) show that CMM closely approximates MNP, which is a reasonable but intractable model for pricing problems. In Online Appendix EC.2.3, we use an example of the single product monopoly pricing problem to numerically illustrate the difference between our model and the robust pricing model based on mean and variance.

3. Bundle Size Pricing

In this section, we formally define the BSP problem. Consider a firm selling n heterogeneous products to a population of homogeneous customers, whose valuations for products are denoted by a random vector $\tilde{\boldsymbol{u}} = (\tilde{u}_1, \ldots, \tilde{u}_n)^T$. Assume that $\tilde{\boldsymbol{u}}$ follows a given distribution F with finite first and second moments and

$\tilde{u}_0 = 0$ is the deterministic valuation for the outside option.

The firm offers a set of bundle sizes denoted by $\mathcal{S}$, where $\mathcal{S} \subseteq [n] = \{1, \ldots, n\}$, and $|\mathcal{S}| = m$. Let $\boldsymbol{p}$ be the price vector of bundles. The firm sets p_s for a bundle of size $s \in \mathcal{S}$ to maximize expected profit, and customers choose bundles (or the outside option) to maximize total surplus, that is, the valuation of a bundle minus its price. For simplicity, we normalize the population to size one and assume that the price sensitivity of all the items are identical and equal to one. Following a common assumption in the bundling literature (Chu et al. 2011), the valuations for items are additive, and the purchased bundle includes at most one of each item.

Let $\tilde{\boldsymbol{w}}$ denote the vector of customers' maximum valuation for each size of bundle. Clearly, $\tilde{w}_s$ is a function of bundle size s and valuation $\tilde{\boldsymbol{u}}$ for $s \in \mathcal{S}$. Denote the ith order statistic of $\tilde{\boldsymbol{u}}$ as $\tilde{u}_{(i)}$, that is, the ith smallest value in $\tilde{\boldsymbol{u}}$, and let $(\tilde{u}_{(1)}, \ldots, \tilde{u}_{(n)})^T$ be the vector of random valuations in increasing order. Then, $\tilde{w}_s(\tilde{\boldsymbol{u}})$ can be written as

$$\tilde{w}_s(\tilde{\boldsymbol{u}}) = \sum_{k=0}^{s-1} \tilde{u}_{(n-k)} \quad \forall s \in \mathcal{S}.$$

This representation means that, for a bundle of size s, customers select s items that they value most. Clearly, $\tilde{w}_0 = \tilde{u}_0 = 0$.

Let c_s denote the cost of the size-s bundle. We assume that the marginal costs are homogeneous among products, which is typically the case in most BSP applications, in particular for information goods. Thus, c_s is a deterministic parameter. Abdallah (2019) shows that the efficiency of BSP decreases as the heterogeneity in marginal costs increases. This also shows that BSP is not a suitable mechanism if the marginal cost differs significantly. In Online Appendix EC.3, we propose two solutions to incorporate the effect of cost heterogeneity. Here, we focus on the homogeneous case.

Because we have already normalized the population to one, the expected demand for a given size of bundle is just the probability of choosing this bundle. Denote the joint distribution of $\tilde{\boldsymbol{w}}$ as G, which is a function of F. Then, the BSP problem can be formulated as follows:

$$\text{(BSP)} \max_{\boldsymbol{p} \geq 0} \sum_{s \in \mathcal{S}} (p_s - c_s) q_s^*(\boldsymbol{p})$$
$$\text{s.t.} \quad q_s^*(\boldsymbol{p}) = \mathbb{P}_{\tilde{\boldsymbol{w}} \sim G}\left(s = \arg\max_{i \in \{0\} \cup \mathcal{S}} \{\tilde{w}_i(\tilde{\boldsymbol{u}}) - p_i\} \right) \forall s \in \mathcal{S}, \tag{1}$$

where $q_s^*(\boldsymbol{p})$ denote the induced expected demand for a bundle of size s given price $\boldsymbol{p}$.

Given distribution G and assuming that there is no inventory constraint, (1) is a standard profit-maximization problem under the discrete choice framework. However, the difficulty of this problem lies in characterizing the distribution G, the expected demand $\boldsymbol{q}^*(\boldsymbol{p})$, and more importantly solving the bilevel program under G, which appears intractable for general F. In the next section, we introduce our approach to construct a tractable approximation to this problem using the cross-moment model.

4. BSP Under the Cross-Moment Model

Though characterizing the distribution G is difficult, the first and second empirical moments of $\tilde{\boldsymbol{w}}(\tilde{\boldsymbol{u}})$ can be easily simulated from distribution F, and these empirical estimates are unbiased. Let $\boldsymbol{\omega} = \mathbb{E}_G[\tilde{\boldsymbol{w}}(\tilde{\boldsymbol{u}})]$ and $\boldsymbol{\Sigma} = \mathrm{Cov}_G(\tilde{\boldsymbol{w}}(\tilde{\boldsymbol{u}}))$, and let Θ be the set of all distributions of $\tilde{\boldsymbol{w}}$ that have the expectation $\boldsymbol{\omega}$ and covariance $\boldsymbol{\Sigma}$, that is, $\Theta = \{\theta \mid \mathbb{E}_\theta[\tilde{\boldsymbol{w}}] = \boldsymbol{\omega}, \mathrm{Cov}_\theta(\tilde{\boldsymbol{w}}) = \boldsymbol{\Sigma}\}$. Clearly, G is an element in Θ. For given price $\boldsymbol{p}$, CMM solves the following upper bound of the expected utility surplus of customers:

$$\text{(CMM)} \quad Z^* = \sup_{\theta \in \Theta} \mathbb{E}_\theta\left(\max_{s \in \{0\} \cup \mathcal{S}} \{\tilde{w}_s(\tilde{\boldsymbol{u}}) - p_s\} \right). \tag{2}$$

More details of CMM and its SDP reformulation are presented in Online Appendix EC.1.1. In this paper, we propose to approximate Problem (1) with

$$\text{(BSP-CMM)}$$
$$\max_{\boldsymbol{p} \geq 0} \sum_{s \in \mathcal{S}} (p_s - c_s) q_s(\boldsymbol{p})$$
$$\text{s.t.} \ q_s(\boldsymbol{p}) = \mathbb{P}_{\tilde{\boldsymbol{w}} \sim \theta^*(\boldsymbol{p})}\left(s = \arg\max_{i \in \{0\} \cup \mathcal{S}} \{\tilde{w}_i(\tilde{\boldsymbol{u}}) - p_i\} \right) \forall s \in \mathcal{S}, \tag{3}$$

where $\theta^*(\boldsymbol{p})$ is the optimal distribution that solves Problem (2), that is,

$$\theta^*(\boldsymbol{p}) = \arg\sup_{\theta \in \Theta} \mathbb{E}_\theta\left(\max_{s \in \{0\} \cup \mathcal{S}} \{\tilde{w}_s(\tilde{\boldsymbol{u}}) - p_s\} \right). \tag{4}$$

Note that, for different values of $\boldsymbol{p}$, $\theta^*(\boldsymbol{p})$ could be different for a fixed Θ. There are two main reasons for us to propose this approximation. First, though $\theta^*(\boldsymbol{p})$ is not necessarily the same as G, they share the first and second moments, and thus, we expect that $\boldsymbol{q}(\boldsymbol{p})$ is a good approximation of $\boldsymbol{q}^*(\boldsymbol{p})$. This conjecture is supported by the numerical experiments in the next section, in which we numerically examine the value of capturing the correlation of valuations. In particular, the approximation error of CMM to the ground truth is much smaller than the approach that ignores correlations. It is also shown in the literature (e.g., Chu et al. 2011) that the correlation structure is crucial for optimizing bundle prices.

The second reason is that computing choice probabilities of CMM is more tractable than other choice models, such as MNP. MNP can also capture correlations among choices, but computing choice probabilities is equivalent to evaluating multidimensional integrals. By contrast, Natarajan and Teo (2017) reduce the size of the SDP formulation of the CMM proposed by Mishra et al. (2012), and Ahipaşaoğlu et al. (2018) show that the choice probabilities under CMM can be computed by maximizing a concave function in a unit simplex under the assumption that the covariance matrix of the customers' valuation for all alternatives (including the outside option) is full rank. Interestingly, Mishra et al. (2012) and Ahipaşaoğlu et al. (2015) show by numerical examples that CMM closely approximates MNP in choice probability predictions.

4.1. Main Result

In the theorem that follows, we show that $\boldsymbol{q}(\boldsymbol{p})$ is the solution to a concave maximization problem in the simplex with the outside option eliminated (the dimension of the simplex is m).

Theorem 1. *In the BSP problem under CMM, assume that $\boldsymbol{\Sigma} > \boldsymbol{0}$.Then, the expected demand is given by*

$$\boldsymbol{q}(\boldsymbol{p}) = \arg\max_{\boldsymbol{x}\in\boldsymbol{\Delta}_m}\left\{(\boldsymbol{\omega}-\boldsymbol{p})^T\boldsymbol{x} + \mathrm{tr}\left(\left(\boldsymbol{\Sigma}^{1/2}\boldsymbol{S}(\boldsymbol{x})\boldsymbol{\Sigma}^{1/2}\right)^{1/2}\right)\right\}, \tag{5}$$

where $\boldsymbol{S}(\boldsymbol{x}) = \mathrm{Diag}(\boldsymbol{x}) - \boldsymbol{x}\boldsymbol{x}^T$, $\boldsymbol{B} = \boldsymbol{A}^{1/2}$ is the unique positive semidefinite square root of a matrix $\boldsymbol{A} \geq \boldsymbol{0}$ such that $\boldsymbol{A} = \boldsymbol{B}^2$, and $\boldsymbol{\Delta}_m \triangleq \{\boldsymbol{x}\in\mathbb{R}^m_+ \mid \boldsymbol{e}^T\boldsymbol{x} \leq 1\}$.

All the proofs of theorems and lemmas are presented in Online Appendix EC.1.

Note that this result extends theorem 1 of Ahipaşaoğlu et al. (2018), but there are several subtle but important differences. First, as mentioned earlier, the valuation of the outside option is deterministically zero (which can be done by subtracting the valuations of all alternatives by the valuation of the outside option), which breaks the full-rank assumption of the covariance matrix (including the outside option) for their result, meaning that their argument in the proof no longer applies. Recall that our definition of $\boldsymbol{\Sigma}$ excludes the outside option. We bridge this gap by proving Lemma EC.1 (see Online Appendix EC.1.2), in which we establish the equivalence of two SDP formulations of CMM (with and without the outside option), and the key is to establish the equivalence of their feasible regions. We find that a standard technique such as the Schur complement is not sufficient and, finally, fill this gap by using techniques from the *positive semidefinite completion* literature.

Second, if we assume that the valuation of the outside option is also random, then directly applying CMM using the formulation in Ahipaşaoğlu et al. (2018) does not necessarily, to the best of our knowledge, lead to a tractable pricing problem. We elaborate on this technical point in Online Appendix EC.4.

We note that $\boldsymbol{\Sigma} > 0$ is satisfied for all the BSP problem cases that we have tested numerically. In unusual cases in which $\boldsymbol{\Sigma}$ is singular, we can simply add a small perturbation $\epsilon\boldsymbol{I}$ (in which $\boldsymbol{I}$ is the identity matrix) to $\boldsymbol{\Sigma}$, and all our results apply.

This theorem is a key step to develop a tractable approximation for the BSP problem. It explicitly characterizes the expected demand $\boldsymbol{q}(\boldsymbol{p})$ under CMM as a function of bundle price. Given this demand, we can write the objective as a function of choice probabilities $\boldsymbol{q}$ rather than directly optimizing over bundle prices $\boldsymbol{p}$. Before doing this, we need to establish their relationship.

Denote $\mathrm{int}(\boldsymbol{\Delta}_m)$ the interior of the simplex $\boldsymbol{\Delta}_m$, that is, $\mathrm{int}(\boldsymbol{\Delta}_m) \triangleq \{\boldsymbol{x}\in\mathbb{R}^m : \boldsymbol{e}^T\boldsymbol{x} < 1, \boldsymbol{x} > \boldsymbol{0}\}$. Let

$$f(\boldsymbol{x}) = \mathrm{tr}\left(\left(\boldsymbol{\Sigma}^{1/2}\boldsymbol{S}(\boldsymbol{x})\boldsymbol{\Sigma}^{1/2}\right)^{1/2}\right).$$

Ahipaşaoğlu et al. (2018) show that $f(\boldsymbol{x})$ is strongly concave over $\boldsymbol{\Delta}_m$. Moreover, if $\boldsymbol{\Sigma} > 0$, then the gradient of $f(\boldsymbol{x})$ in $\mathrm{int}(\boldsymbol{\Delta}_m)$ is given by

$$\nabla f(\boldsymbol{x}) = \frac{1}{2}\mathrm{diag}\left(\boldsymbol{\Sigma}^{1/2}(\boldsymbol{T}(\boldsymbol{x}))^{-1/2}\boldsymbol{\Sigma}^{1/2}\right) - \boldsymbol{\Sigma}^{1/2}(\boldsymbol{T}(\boldsymbol{x}))^{-1/2}\boldsymbol{\Sigma}^{1/2}\boldsymbol{x},$$

where $\boldsymbol{T}(\boldsymbol{x}) = \boldsymbol{\Sigma}^{1/2}\boldsymbol{S}(\boldsymbol{x})\boldsymbol{\Sigma}^{1/2}$.

In the next result, we show that the optimal solution $\boldsymbol{x}^*$ to (5) must lie in $\mathrm{int}(\boldsymbol{\Delta}_m)$ provided that $\boldsymbol{\Sigma}$ is nonsingular.

Lemma 1. *Assume that $\boldsymbol{\Sigma} > 0$. Then, $\|\nabla f(\cdot)\|$ approaches infinity as $\boldsymbol{x}$ approaches the boundary of $\boldsymbol{\Delta}_m$.*

Based on Lemma 1, the first-order optimality condition for Problem (5) is $\boldsymbol{p} - \boldsymbol{\omega} = \nabla f(\boldsymbol{q})$, which gives the unique optimal choice probabilities for any given $\boldsymbol{p}$. The next result establishes the relationship between $\boldsymbol{p}$ and $\boldsymbol{q}(\boldsymbol{p})$.

Lemma 2. *Assume that $\boldsymbol{\Sigma} > 0$, and let $H : \mathbb{R}^m \longmapsto \mathrm{int}(\boldsymbol{\Delta}_m)$ be the mapping from $\boldsymbol{p}$ to $\boldsymbol{q}$ in CMM. Then, H is a bijection (one-to-one correspondence).*

The one-to-one correspondence implies that CMM can generate all the choice probabilities in $\mathrm{int}(\boldsymbol{\Delta}_m)$, which is crucial in solving the BSP problem. Given the one-to-one correspondence, we can express the price vector $\boldsymbol{p}$ as a function of choice probability $\boldsymbol{q}$, that is, $\boldsymbol{p} = \boldsymbol{\omega} + \nabla f(\boldsymbol{q})$, and then optimize profit over $\boldsymbol{q}$. Therefore, the BSP-CMM Problem (3) can be equivalently written as follows:

$$\max_{\boldsymbol{q}\in\boldsymbol{\Delta}_m} \pi(\boldsymbol{q}) = \boldsymbol{q}^T(\boldsymbol{\omega}-\boldsymbol{c}) + \frac{1}{2}\boldsymbol{q}^T\mathrm{diag}\left(\boldsymbol{\Sigma}^{1/2}(\boldsymbol{T}(\boldsymbol{q}))^{-1/2}\boldsymbol{\Sigma}^{1/2}\right) - \boldsymbol{q}^T\boldsymbol{\Sigma}^{1/2}(\boldsymbol{T}(\boldsymbol{q}))^{-1/2}\boldsymbol{\Sigma}^{1/2}\boldsymbol{q}, \tag{6}$$

where $\boldsymbol{c}$ is the vector of the expected cost of bundles and $\pi(\boldsymbol{q})$ is the expected profit in BSP-CMM.

The next theorem shows that solving the BSP problem is surprisingly easy under CMM as long as we are given the mean vector and covariance matrix of customer valuations.

Theorem 2. *Assume that* $\boldsymbol{\Sigma} \succ 0$. *Then, the objective function* $\pi(\boldsymbol{q})$ *in* Problem (6) *is concave.*

It is known in the literature that computing choice probabilities of CMM with given model parameters is equivalent to solving a convex program. However, it is not obvious whether the pricing problem under CMM is also tractable given the complicated form of $\pi(\boldsymbol{q})$. In Theorem 2, we give a positive answer. We prove this result by checking the convexity of the epigraph of $\pi(\boldsymbol{q})$ and exploiting the properties of some matrix functions. Other than solving the BSP problem, another implication of Theorem 2 is that we can use CMM to approximately solve a series of bundle pricing problems in a unified way. The key step is to transform these problems into a multidimensional pricing problem. Then, we just need to solve Model (6) after getting moments.

Recall that we assume homogeneous price sensitivity for all products. Then, a natural question is whether the convex property extends to the case of heterogeneous price sensitivity. Unfortunately, this is not true. Two difficulties arise in this setting. First, we can no longer transform the BSP into a multiproduct pricing problem because the price sensitivity of a given size of bundle differs for different customers. Second, even if one can obtain a constant but heterogeneous price sensitivity for each bundle size, the pricing problem in Theorem 2 is not necessarily convex. However, there might exist some numerical solutions to this issue, which we leave for future study.

Given the offer set $\mathcal{S}$, there are three relevant parameters in solving Problem (6): the mean vector $\boldsymbol{\omega}$, the cost vector $\boldsymbol{c}$, and the covariance matrix $\boldsymbol{\Sigma}$. We can obtain the empirical estimates of the two moments by first drawing enough samples of valuations for each product from F and then sorting the samples in decreasing order and computing the sample mean and covariance matrix of the cumulative sums. In Online Appendix EC.5, we discuss how to use the *method of simulated moments* to estimate F or directly estimate $\boldsymbol{\omega}$ and $\boldsymbol{\Sigma}$ from real data. Because the product cost is homogeneous, vector $\boldsymbol{c}$ can be easily obtained by multiplying the bundle size with marginal cost.

It is worth mentioning that the extremal distribution underlying CMM can be constructed as a mixture of normals. As mentioned earlier, a series of numerical examples (see, e.g., Mishra et al. 2012, Ahipaşaoğlu et al. 2015) show that CMM serves as a close approximation to the MNP model but is more tractable. This implies that, whenever MNP is a good approximation to the ground truth, CMM gives good approximations too. It is well known that MNP is one of the most general among parametric choice models (Train 2009). However, it is largely ignored by the revenue management community because even computing choice probabilities of MNP requires evaluating multidimensional integrals, which is computationally expensive. In this sense, we believe that CMM is a competitive alternative to MNP for pricing problems.

Note that it is hard to show the closeness of the numerical performance of CMM and MNP for the general BSP problem. However, this comparison is possible in the special case of a single dimension. In the following, we first study the single-size BSP under CMM; we then numerically compare CMM and MNP in this much simplified setting.

4.2. Single Size Pricing Under CMM

In practice, firms may only provide a few bundle sizes for customers. In this case, CMM is flexible in optimizing any given bundle sizes. When there is only one size to optimize, Problem (6) reduces to the following simple, one-dimensional nonlinear equation.

Corollary 1. *If* $\mathcal{S} = \{i\}$ *for* $i \in [n]$, *then* (6) *is equivalent to solving the following equation:*

$$\omega_i - c_i + \frac{\sigma_i(4x^2 - 6x + 1)}{4(1-x)\sqrt{x - x^2}} = 0, \tag{7}$$

where x *is the choice probability,* ω_i *and* σ_i^2 *are the mean and variance of the valuation of a bundle of size* i, *and* c_i *is the expected cost.*

This result is a direct consequence of Lemma 1 and the first-order condition of (6), so we omit the proof here. By solving this problem for any size, one can obtain the optimal single-size pricing policy under CMM.

We can also write the one-to-one correspondence between price and demand under CMM as

$$p_{\mathrm{CMM}} = \omega + \frac{\sigma(1-2x)}{2\sqrt{x - x^2}},$$

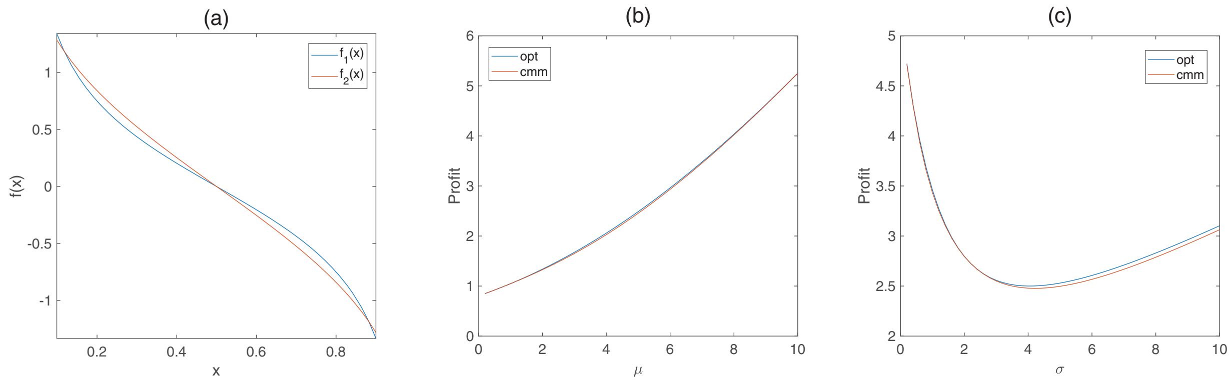
where we omit the subscript for ease of illustration. Similarly, it is easy to show that the relation of price and demand under MNP is given as

$$p_{\mathrm{MNP}} = \omega + \sigma\,\Phi^{-1}(1-x),$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function.

Intuitively, for the single-dimensional pricing problem, CMM uses the function $f_1(x) = (1-2x)/(2\sqrt{x-x^2})$ to approximate $f_2(x) = \Phi^{-1}(1-x)$, which has a more complex form. We plot these two functions in Figure 1(a). We can see that they are very close to each other. To see how CMM approximates MNP in this setup, in Figure 1(b) and (c), we first compute the optimal profit by assuming MNP as the ground truth and then compute the

Figure 1. (Color online) Comparison of CMM and MNP for Single-Size BSP



Notes. (a) Function $f_1(x) = (1-2x)/(2\sqrt{x-x^2})$ in CMM and function $f_2(x) = \Phi^{-1}(1-x)$ in MNP. (b) The optimal profit of the single-size BSP with MNP as ground truth and the profit under prices by CMM. $\sigma = 5, \mu \in [0, 10]$. (c) The optimal profit of the single-size BSP with MNP as ground truth and the profit under prices by CMM. $\mu = 5, \sigma \in (0, 10]$.

optimal prices under CMM using the same moments. We finally compute the resulting profits obtained by the prices given by CMM under the demand function of MNP. Still, the profits obtained by CMM are very close to the true optimal profit under MNP.

5. Numerical Experiments

In this section, we investigate the numerical performance of the pricing solutions of CMM via simulations. We first examine the value of capturing correlations of valuations in prediction of choice probabilities under BSP. We then compare our approach with two state-of-the-art heuristics, the simulation optimization (SO) method proposed by Chu et al. (2011), and the closed-form heuristic solution proposed by Abdallah et al. (2021). The experiments were run on a laptop with an Intel(R) i7-8550U CPU (2.0 GHz) with 16 GB RAM. All the problem instances under CMM are solved using "fmincon," which is a nonlinear optimization solver in MATLAB R2018a.

5.1. The Value of Capturing Correlation

In this section, we use simulation to illustrate the importance of capturing correlations between valuations of bundles for given bundle size prices. To this end, we compare the approximation error of CMM with another choice model that completely ignores correlations but captures full marginal distribution, denoted by G_s, for bundle size $s \in \mathcal{S}$. We assume that the joint distribution G for bundles is given by $G = \prod_{s\in\mathcal{S}} G_s$ under this model (called the independent model, simplified as IM).

We measure the approximation error using *total variation distance*, which is one of the most common statistical distance measures. Consider two discrete probability distributions P and Q supported on $[K]$. The total variation distance between P and Q is defined as

$$\delta(P, Q) = \frac{1}{2}\sum_{k\in[K]} |P(k) - Q(k)| \,.$$

See chapter 4 in Levin and Peres (2017) for more details.

The basic setup is as follows: for ease of illustration, we set $n = 5$ and $\mu_i = \mu = 3$ for $i \in [n]$. We consider both independent and correlated valuations for products. Note that, even if valuations for products are independent, the valuations for bundles are likely to be highly correlated. For the independent case, the marginal distributions include exponential$(1/\mu)$, uniform$(0, \mu)$, logit$(\mu, 1)$, lognorm$(\mu/3, 0.25)$, and normal$(\mu, 1)$. For the correlated case, we set F as normal$(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with randomly generated $\boldsymbol{\Sigma}$, of which the eigenvalues are independently and uniformly sampled from $[0, 1]$, and the eigenvectors are generated as the orthogonal basis for the range of a random matrix with independent and identically distributed entries uniformly sampled from $[-1, 1]$.

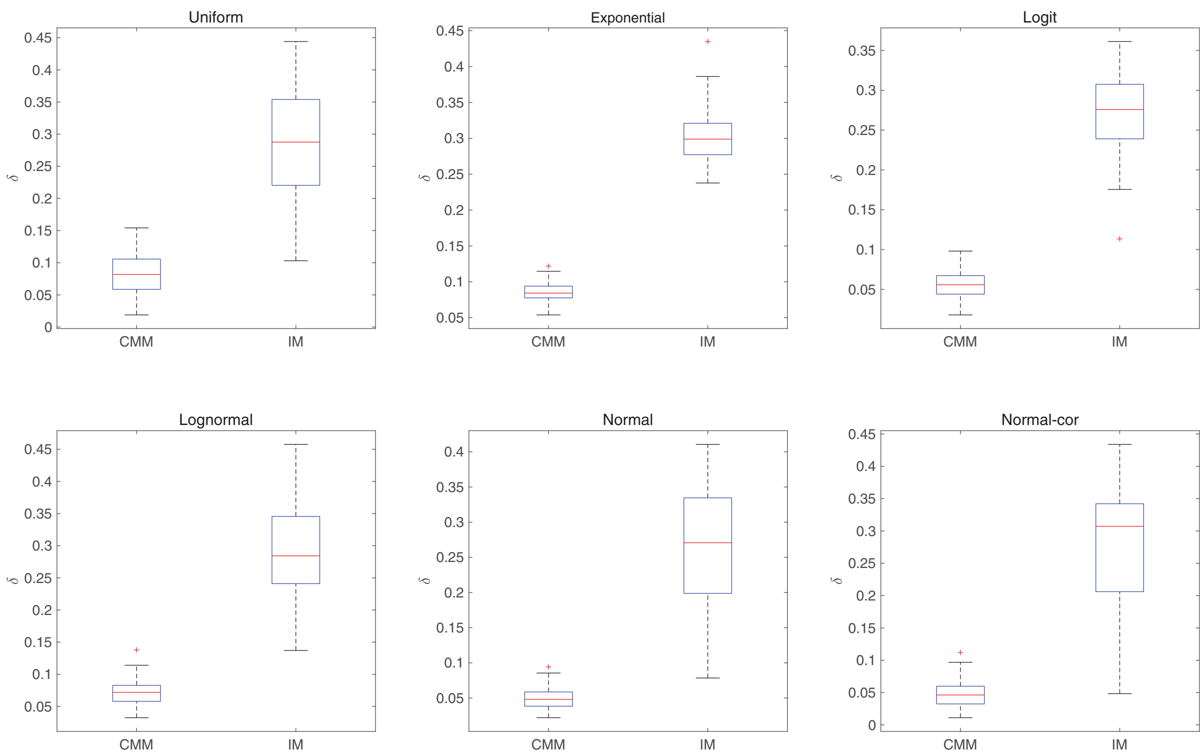
We also set feasible bundle prices p_s randomly. By feasible, we mean p_s is monotonically increasing in bundle size s, and the average price per product (i.e., p_s/s) is monotonically decreasing in s. Specifically, we first sample n numbers uniformly from $(\nu - 0.5, \nu + 0.5)$, where ν is the mean of the marginals, and then sort these numbers in decreasing order. We use these numbers as average per-product prices for bundles and obtain bundle prices by multiplying the per-product prices with bundle sizes. The details of the simulations are as follows.

First, we draw samples from F, which is the ground-truth joint distribution of valuations over products. We calculate corresponding valuations for each size of bundle ω_s for $s \in \mathcal{S}$. Using the samples and feasible prices, we can compute an empirical estimate of the choice probabilities of bundles $\hat{\boldsymbol{q}}(\boldsymbol{p}_s)$, which we set as ground truth.

Second, we calculate the sample moments of valuations for bundles $\hat{\boldsymbol{\omega}}$ and $\hat{\boldsymbol{\Sigma}}$ as inputs of CMM and then compute the choice probabilities under CMM by solving Problem (5).

Third, we estimate the empirical marginal distributions of valuations for each size of bundle, denoted by

Figure 2. (Color online) Comparison of Total Variation Distances $\delta(P_{\mathsf{cmm}}, Q_t)$ and $\delta(P_{\mathsf{im}}, Q_t)$



$\hat{G}_s$ for $s \in \mathcal{S}$, and set $G = \prod_{s\in\mathcal{S}} \hat{G}_s$. Using these marginals, we simulate customer valuations for bundles independently and calculate their choices based on the prices p_s, which give us empirical estimates of choice probabilities under IM.

Finally, we compare the total variation distances between choice probabilities generated from these three models. Specifically, let P_{cmm} and P_{im} denote the choice probabilities under CMM and IM and let $Q_t = \hat{\boldsymbol{q}}(\boldsymbol{p}_s)$ denote the ground truth. In each simulation, we calculate both $\delta(P_{\mathsf{cmm}}, Q_t)$ and $\delta(P_{\mathsf{im}}, Q_t)$. For each distribution, we run 100 simulations, and for each simulation we draw 10,000 samples from F. We allow free disposal in the simulations; that is, we assume that a customer has zero utility for any negative-valued product in the bundle the customer purchased.

We plot $\delta(P_{\mathsf{cmm}}, Q_t)$ and $\delta(P_{\mathsf{im}}, Q_t)$ in the box plot in Figure 2 for each distribution. We can see that $\delta(P_{\mathsf{cmm}}, Q_t)$ is much smaller than $\delta(P_{\mathsf{im}}, Q_t)$ in all cases. This partially validates the value of capturing correlations between bundles in predicting choice probabilities. In particular, $\delta(P_{\mathsf{cmm}}, Q_t)$ is less than 0.15 in almost all cases and is less than 0.1 under logit and normal. This implies that CMM fit the choice probabilities under these distributions quite well.

5.2. Comparison with Existing Approaches

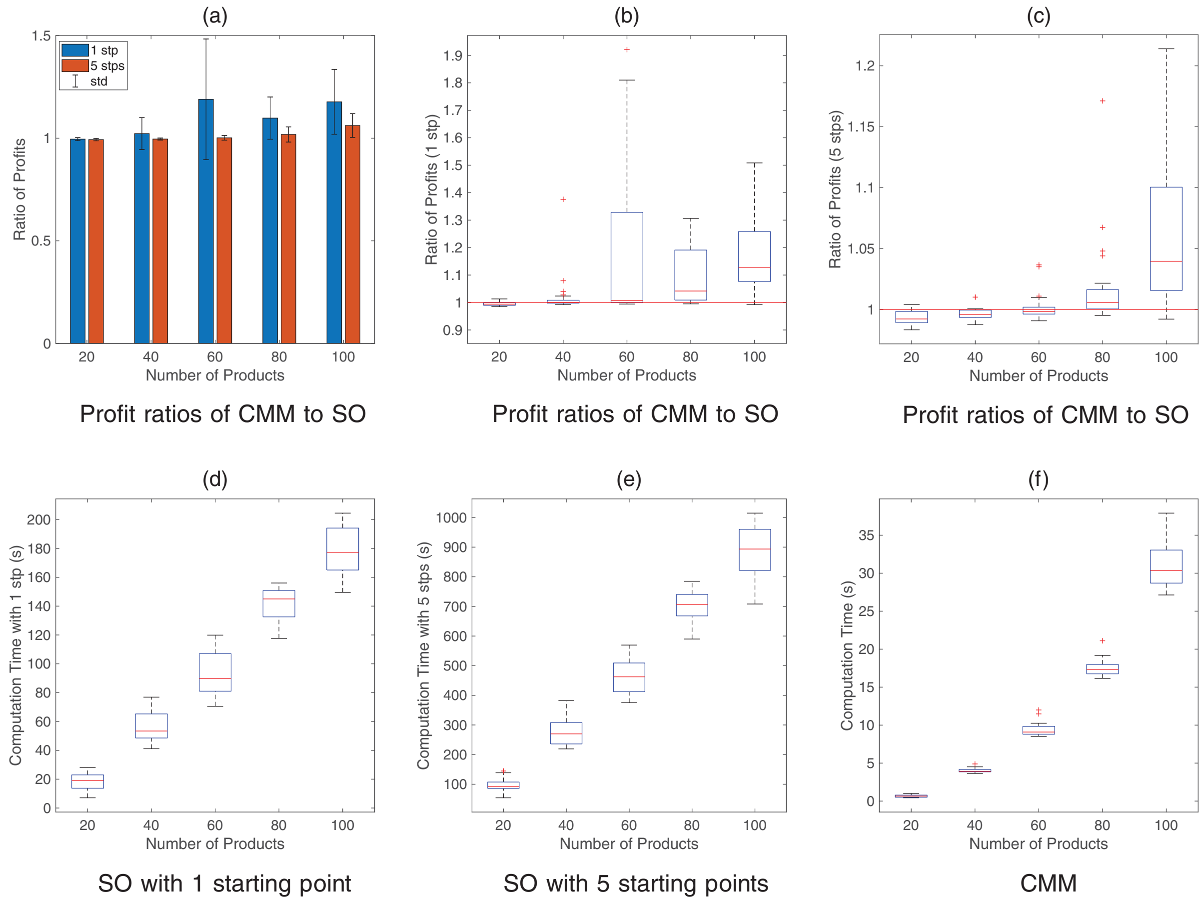
In this section, we compare CMM with two state-of-the-art benchmarks, Chu et al. (2011) and Abdallah et al. (2021), for the BSP problem. Following Abdallah et al. (2021), the basic experiment setup is as follows: the marginal distributions of valuation of all products are set as the same, including exponential($1/\mu$); uniform($0, \mu$); truncated logit($\mu, 1$) supported on $(0, \infty)$; truncated normal($\mu, 1$) supported on $(0, \infty)$, which are independent over products; and truncated normal($\mu, 1$) supported on $(0, \infty)$ with positive and negative correlations, which are parametrized by Gaussian copula. In the correlation matrix of Gaussian copula, the proxy is set to $\rho = 1$ so that the diagonal element is one and the off-diagonal elements are $\pm 1/n$. Denote γ as the cost ratio and let $\boldsymbol{c} = \gamma \cdot \boldsymbol{\nu}$, where $\gamma \in \{0, 0.3, 0.5, 0.8\}$, and ν is the mean of marginals.

The numerical experiment is performed as follows: we first compute optimal bundle pricing policies of both CMM and benchmark approaches (see Online Appendix EC.2 for details) under each parameter setting and then use these pricing policies to calculate out-of-sample mean profits using simulation, where we generate 10,000 customer valuations for each case. We finally compute the mean profit attained under each policy and the ratio of profits of CMM to that of benchmarks. CMM outperforms the benchmarks if the ratio is greater than one and vice versa.

5.2.1. Comparison with Chu et al. (2011).

We consider the number of products $n \in \{20, 40, 60, 80, 100\}$, and uniformly sample μ from $(1, 5)$ for each n. Because the

Figure 3. (Color online) Comparison of CMM and Chu et al. (2011)



SO approach is local in nature, the obtained solution highly depends on the starting point (recall that the BSP problem is not necessarily convex). In this experiment, we use a random starting point, at which the price of each bundle size is initialized as the mean valuation (i.e., $\boldsymbol{\omega}$, obtained by simulation) times a random number uniformly sampled from (0.5, 1.5). We then compute two sets of solutions to the BSP problem. One uses a single starting point, and the other uses five starting points and then we pick the solution with the highest objective value. For CMM, we just uniformly sample a starting point (demand) from $\boldsymbol{\Delta}_m$.

We also tried another set of starting points based on the marginal cost of each size of bundle. The results under these starting points are inferior to those under the starting points based on mean valuations as reported in this section. See Online Appendix EC.2.1 for more details.

We summarize the computational results including the ratio of out-of-sample profits of CMM to Chu et al. (2011) and computation times in Figure 3. In Figure 3(a), the bar plots correspond to the mean ratios of CMM to the benchmark with blue bars corresponding to results using a single starting point and orange bars corresponding to five starting points. The error bars represent the standard deviations of the ratios. Figure 3, (b) and (c), shows box plots of profit ratios with one starting point and five starting points, respectively, and the red reference lines correspond to the ratio of one (the same interpretation in the following figures).

We can see that the performance of the solutions with five starting points is consistently better than those with one starting point. Also, the performance of the two methods is very close when n is small (the SO with five starting points slightly outperforms CMM, and the advantage is within 2%) although CMM outperforms the SO approach when n increases. The advantage of CMM is less significant for the case of five starting points compared with the case of one starting point. This is intuitive because searching for more starting points most likely improves the solution of SO. Also, as n increases, the increasing standard deviation of the profit ratios illustrates that the SO approach is highly sensitive to the choice of starting points.

Figure 3, (d)–(f), displays box plots of computation times of the two approaches. We observe that the computation times (note that the scales of the vertical axes in Figure 3, (d)–(f), are different) of SO with five

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starting points are around five times of those with one starting point, and both of these two cases are much more computationally expensive than CMM.

To summarize, the success of CMM compared with the SO approach is when the number of products increases. This is not surprising. On the one hand, CMM can effectively capture the correlation of valuations over bundles using moments, and the approximation error is minor as shown by the numerical results in the last section. On the other hand, because the BSP problem under CMM is convex, we can always solve to optimality efficiently for reasonable problem sizes. The good approximation together with guaranteed optimality imply that CMM can produce competitive prices. However, the SO approach can get stuck at the local optimum with high probability when n is large. This explains why the performance of CMM stands out as n increases.

5.2.2. The Value of Complexity of the Pricing Policy. In this section, we numerically investigate the value of offering full bundle sizes over the single-size pricing policy under CMM. We consider the number of products $n \in \{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$. For the single-size pricing policy, we first solve Problem (7) for every bundle size $i \in [n]$ and then pick the bundle size and its price that attains the largest objective value. We plot the ratio of profits in Figure 4, in which we group the values based on cost ratio in the left box plot and group based on number of products in the right one. We see that the advantage of the full-size policy over the single-size policy increases in cost ratio, meaning that it's better to offer full bundle sizes if the cost is high. Also, the advantage decreases in the number of products. This observation coincides with the result of Abdallah et al. (2021), who show that offering a single bundle size is asymptotically optimal for the BSP problem.

5.2.3. Comparison with Abdallah et al. (2021). We compare both the full- and single-size pricing policies under CMM with the closed-form heuristic in Abdallah et al. (2021). For the full-size policy of CMM, we consider the number of products $n \in \{5, 10, 50, 100, 200, 300\}$, and for the single-size policy of CMM, we consider larger scales in which $n \in \{5, 10, 50, 100, 500, 1000\}$. Also, we fix $\mu = 3$. The computational results are summarized in Figure 5.

From Figure 5, we see that both the full- and single-size policies of CMM outperform Abdallah et al. (2021) in most cases. The advantage is more significant when n is small, say $n \leq 50$, especially when the marginal cost is high. The reason is that the full-size policy of CMM is more flexible than the single-size policy of Abdallah et al. (2021) in extracting customer surplus when the cost is high.

The advantage of CMM decreases as n increases; in particular, the benchmark slightly outperforms the full-size CMM in nearly half of the total instances when $n = 300$ as shown in Figure 5(b). However, the single-size CMM performs consistently better than the benchmark. The reason is that pure bundling approaches optimal when n is large, cost is low, and valuations for products are independent (these cases account for nearly half of the total instances tested). In such cases, the optimal solution to the BSP problem is nearly degenerate, and CMM with full size returns only an approximate solution, whereas the single-size solution deals with this degeneracy issue directly.

For large-scale BSP, Abdallah et al. (2021) already show that their policy is near optimal. Nonetheless, we note that the single-size policy of CMM still performs well for large-scale problems, and both approaches are computationally efficient; CMM corresponds to root finding of a nonlinear equation with a single variable, and Abdallah et al. (2021) evaluates closed-form expression. In the results of the single-size policy, we

Figure 4. (Color online) The Ratio of Profits of the Full- and Single-Size Policies Under CMM

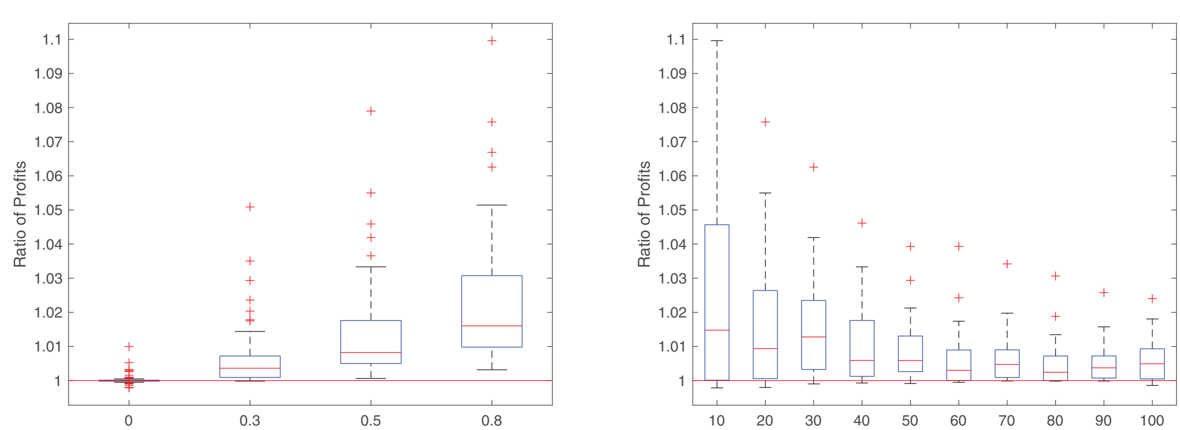
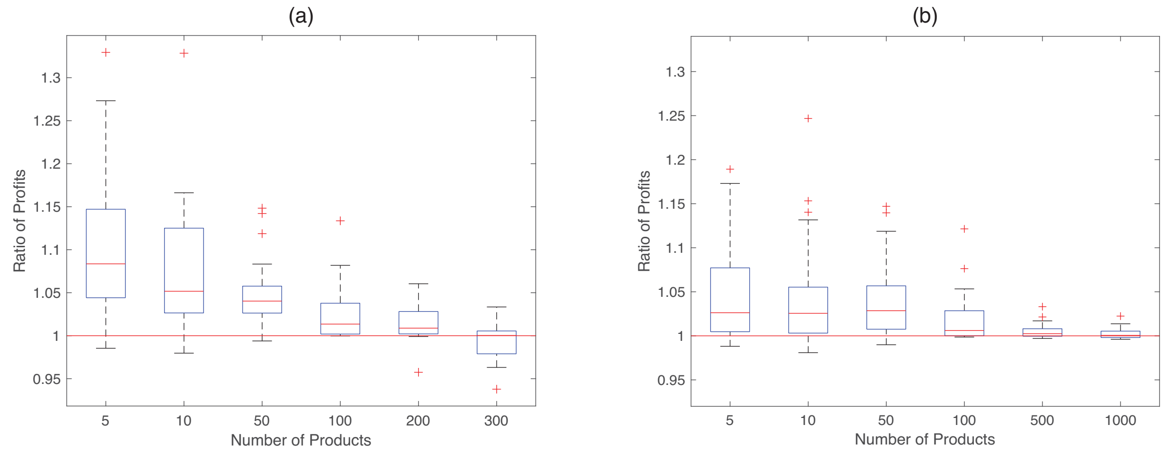


Figure 5. (Color online) The Ratio of Profits of CMM to Abdallah et al. (2021)



Full size CMM to Abdallah et al. (2021)

Single size CMM to Abdallah et al. (2021)

also note that the two methods recommend different but very close bundle sizes and prices. Theoretically, the similarity between these two approaches is that both standard deviation of valuation and cost appear in the expressions (see Online Appendix EC.2.2 for details). However, CMM uses mean valuations of bundles, and Abdallah et al. (2021) uses marginal distribution function of valuation. Moreover, the bundle size in CMM is n (pure bundling) when cost is zero and decreases as cost increases.

6. Conclusion

In this paper, we study the computational aspect of the BSP problem. We propose to solve this problem under CMM, which captures the correlations of valuations over bundles through the first two moments instead of exact distribution functions. We show that the BSP problem under CMM is convex, thus enabling efficient computation of bundle prices. Numerical results demonstrate that our approach performs very well compared with several state-of-the-art methods. The encouraging empirical evidence combined with the capability of capturing correlations provide a unified and efficient approach for solving the BSP problem.

We also provide a few insights regarding the BSP problem and CMM. First, the BSP problem can be converted into a multichoice pricing problem with correlations of valuations, and it is crucial to capture such correlations to construct good approximate solutions. Second, using only moment information in the way of CMM can be reasonable for constructing a good approximation to the BSP problem. Third, CMM is a close approximation to the MNP model, which can capture arbitrary correlations over choices. However, the pricing optimization problem under MNP is intractable in general. In this sense, CMM serves as a tractable alternative to MNP and successfully captures the correlations in the BSP problem. As bundling strategy is pervasive and important in practice, we hope that CMM opens an avenue for solving a wide array of bundle pricing problems, such as the mixed bundling problem.

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